Column vectors

1

a $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$

b $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

2

a $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

b $\begin{bmatrix} 3 \\ -3 \end{bmatrix}$

c $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

3

a

i $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$

ii $\begin{bmatrix} 6 \\ -4 \end{bmatrix}$

b

i $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$

ii $\begin{bmatrix} -5 \\ -3 \end{bmatrix}$

4

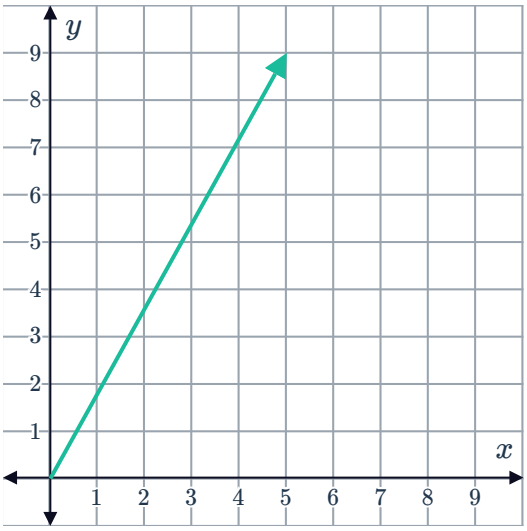
a $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$

b $\begin{bmatrix} -7 \\ 3 \end{bmatrix}$

c $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$

d $\begin{bmatrix} 10 \\ 1 \end{bmatrix}$

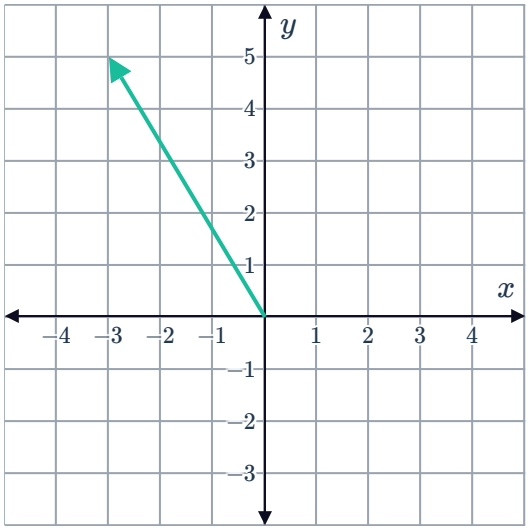
5



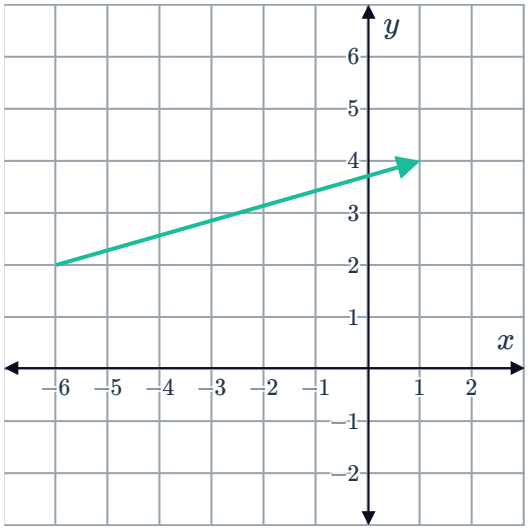
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6

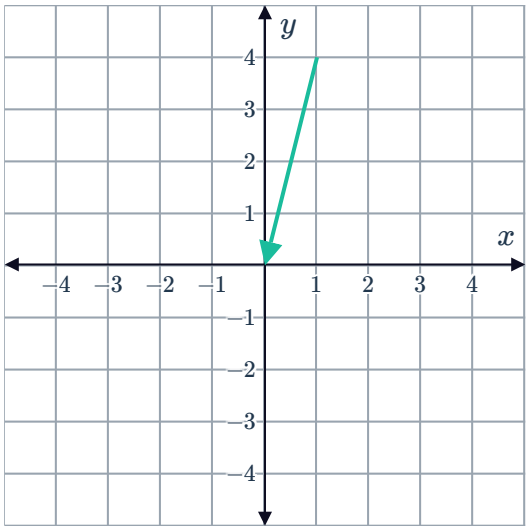
a



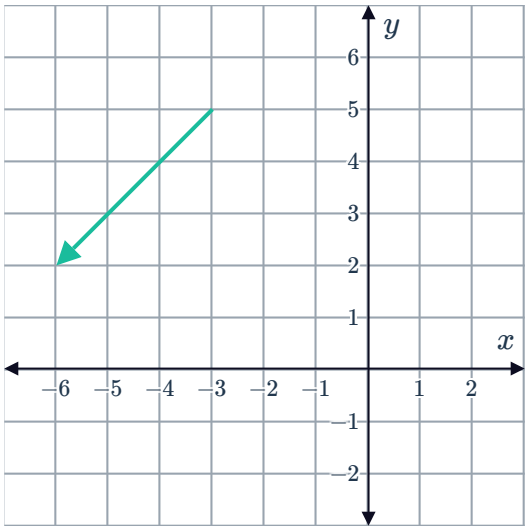
b



c



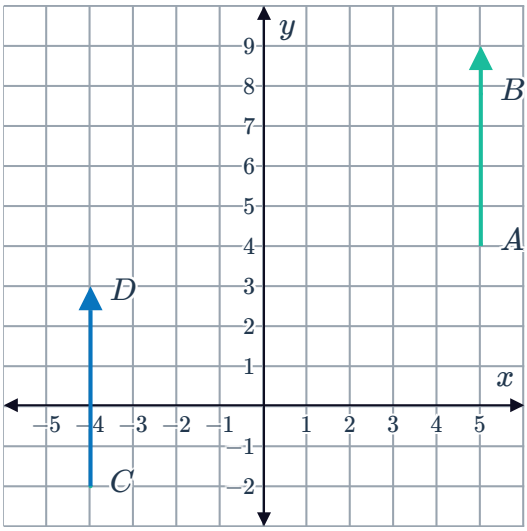
d



7

a

i

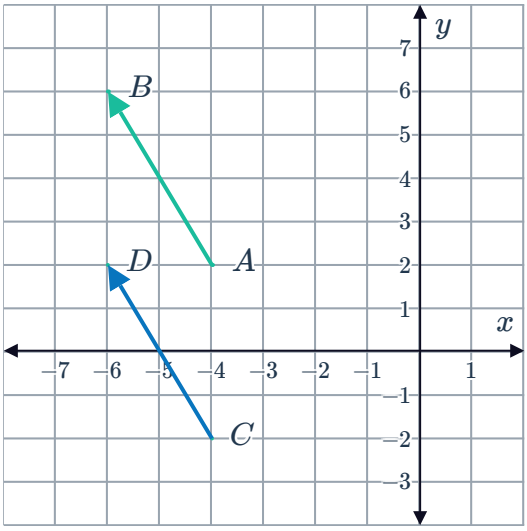


ii Yes

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b

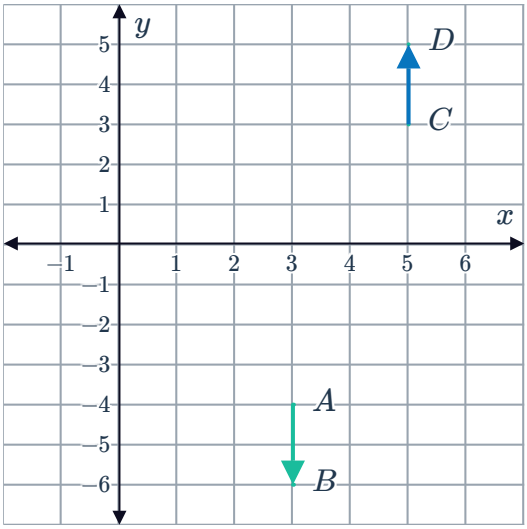
i



ii Yes

c

i



ii No

8

a Yes **b** Yes **c** No **d** No

9 800 km north

Magnitude of a vector (Extended)

10 5 units

11 5 units

12 6 units

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13

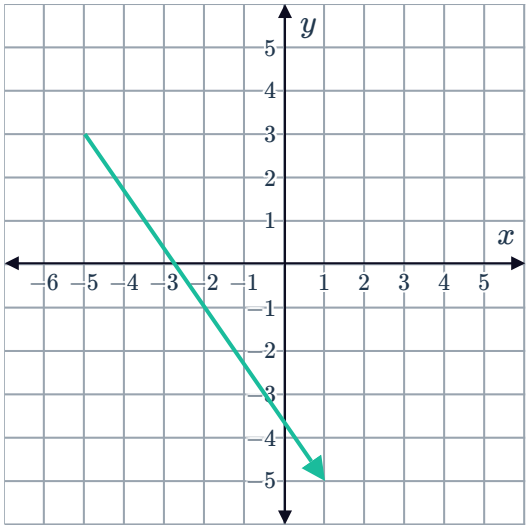
a 15 units

b 51 units

c No, because the displacement is just the direct distance from the beginning point to the end point of the path.

14

a



b 10 units

15 True

16

a 7 units

b $\sqrt{74}$ units

c 17 units

d 17 units

e 6 units

f 6 units

17 $4\sqrt{5}$ units

18 13 units

19 $x = 3, -3$

20 20 km

21 13km

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